University of Glasgow

James Watt School of Engineering

Degrees of MEng, BEng, MSc and BSc in Engineering

ENG1064 Microelectronic Systems 1

Wednesday, 10 May 2023, Release time: 09:30-11:30

Exam duration: 1 hour 30 minutes to complete exam plus 30 mins to upload submission.

Attempt ALL questions.

TOTAL MARKS AVAILABLE
50

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

A calculator may be used. Show intermediate steps in calculations.

Your lecture notes can be used during the exam.

Various types of handwriting might be very challenging for us to read and mark. Where appropriate, please type in your answers.

OVER

Q1

Embedded Systems

Most of the electronic devices that we buy nowadays are based on microcontroller electronics and are built to continuously perform a particular task.

a) Briefly explain what an embedded system is. Detail the main functionalities of such a system and its differences from a personal computer. [3]

b) Below, Figure Q1, shows an example of a modern smart fridge. This is an embedded system with many of its functionalities based on microcontroller electronics.

   i. Detail all the embedded functionalities of a conventional fridge (not smart), i.e. what are the inputs, outputs and what information is being processed by a conventional fridge's microcontroller. (Use a maximum of 200 words) [3]

   ii. Expand your previous answer to describe all the embedded functionalities of the modern smart fridge (i.e. phone connectivity), shown below. (Use a maximum of 200 words) [4]

   iii. You are an embedded systems developer working for a big fridge manufacturer. Your company is developing the next generation of smart fridges and they are looking for a novel fridge functionality that the company will incorporate in their next generation fridges.

   Develop and compose a new, smart feature for a fridge. This can be an imaginative feature. Make all the necessary links to embedded functionalities, i.e. what are the input/output signals, and information is being processed. (Use a maximum of 300 words) [3]

Figure Q1: A modern smart fridge

Continued overleaf

Q2

ADC (Analog-to-digital conversion)

Modern smart watches have many embedded functionalities. One of these functionalities is to accurately measure the **heart rate** of the person wearing the smart watch. The method for measuring the heart rate of a human is outlined below, in Figure 2. As shown in the Figure, there are **green LEDs** that illuminate green light on the skin of person wearing the watch. In addition to the LEDs there are also **sensors** (photodiodes) that are used to detect the green light bouncing off the skin (reflected light). When the heart beats, the intensity of the green light returning back to the sensor is less than the in-between heart beats.

a) A Pulse Width Modulation (PWM) signal is used to illuminate the green LEDs. Describe what an appropriate PWM frequency for turning ON/OFF the green LEDs would be and justify your answer. Note that in humans the average heart rate is 60-100 beats per minute. [3]

b) The green light that is bouncing (reflected) off the person wearing the watch is detected by a light sensor (photodiode). This is an analogue sensor that returns a voltage from 0V to 1.0V, when there is no green light or max green light detected respectively.
This analogue signal is sent to a **12-bit ADC** with 3.3V reference voltage and 0V offset that is used to digitise the signal for further processing.

   i. What is the ADC resolution expressed in milli-volts (mV)? [1]

   ii. What is the range of this ADC? [1]

   iii. What is the quantisation error? [1]

   iv. The output voltage from the green light sensor is 0.93 Volts. What is the digital signal returned by the ADC? [2]

   v. This particular ADC can digitize signals with a rate of 300 kHz. What is the maximum input signal frequency you can send into the ADC? Explain why this limit exists. [2]

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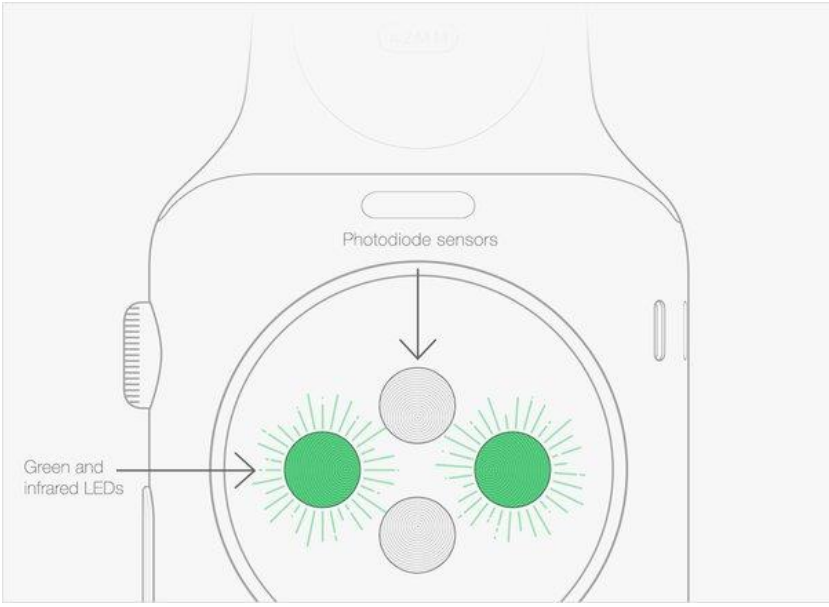

Figure Q2: Smart watch with a heart rate detection capability

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Q3

MCU inputs and Outputs

The Nucleo microcontroller has several pins that can be configured as Digital Outputs. Each of these pins can provide a DC voltage of 3.3 Volts.

a) Summarise how you would connect a switch to the microcontroller [2]

b) Describe in detail, using appropriate diagrams, how you would connect a red LED to one of the MCU pins. The LED operates with a voltage of **2.2 Volts** and **5 mA** current. Label all the components and calculate their values if necessary. [3]

c) Explain if it is possible to connect a second red LED to the same port. Illustrate with a new schematic how you would connect it and label the required components accordingly. [5]

d) You would like to connect a motor to one of the PWM (Pulse Width Modulation) outputs of your microcontroller. For normal operation, the motor requires a voltage of 24 Volts. Explain how you would connect the motor, using appropriate diagrams. [5]

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Q4

I2C communication protocol

Working as an Embedded Systems Developer at a huge fridge manufacturing company, you are looking for your next big challenge. One day, you are asked by your team leader to replace the old analogue fridge temperature sensors with a new digital sensor. You immediately decide to use the **TC74-A0** temperature sensor, as you are familiar with it and used it in the past during your engineering studies at the University of Glasgow.

a) Draw a schematic of how you would connect this sensor to your microcontroller, detailing all the components you would use. [2]

b) In an I2C communication protocol, we usually send data in packets of 8-bits (1-byte). Explain how the microcontroller knows that the data have been received by the temperature sensor? [2]

c) According to the manufacturer, the **address** of this sensor is **0x9**. You would like to read the temperature from this sensor and for that you **send the command 0x0**, while the I2C communication frequency is set at the recommended 100 KHz.

   Draw a diagram that illustrates the information sent through the Data and Clock wires. Give details on each segment of the communication and put appropriate labels on the Voltage/Time axes. [8]

End of question paper